

Instructional Program Design Portfolio

Automotive Systems Fundamentals — A Structured Learning Program

Portfolio evidence demonstrating end-to-end learning/program design capability

Prepared for: Training & Development Specialist application

1. Program Overview

This portfolio presents a complete, structured learning program designed to introduce learners to core automotive systems. It demonstrates the full instructional pathway from identifying learning outcomes through content, activities, practical work, assessment, and completion criteria.

Program title	Automotive Systems Fundamentals
Target audience	Beginner-level vocational/technical learners
Delivery format	Blended: instructor-led learning, demonstrations, discussion and practical activities
Suggested duration	5 modules / approximately 15 learning sessions
Program goal	Build foundational understanding of major automotive systems and basic diagnostic thinking

2. Program Learning Objectives

- Identify the purpose and major components of common automotive systems.
- Explain, in simple technical terms, how major vehicle systems operate and interact.
- Apply workshop safety procedures and appropriate technical vocabulary.
- Carry out basic inspection and fault-identification activities using a structured process.
- Demonstrate learning through quizzes, practical tasks, assignments and a final assessment.

3. Program Structure & Learning Pathway

Module	Focus	Learning Activities	Assessment	Expected Outcome
1	Automotive Fundamentals & Safety	Interactive lesson, component identification, safety demonstration	Short quiz + safety checklist	Learner explains basic vehicle systems and follows workshop safety procedures.
2	Engine System	Diagram analysis, demonstration, guided practical inspection	Component-identification task + quiz	Learner identifies key engine components and explains their basic functions.
3	Electrical System	Circuit demonstration, guided troubleshooting, discussion	Practical circuit task + troubleshooting questions	Learner understands basic vehicle electrical flow and can follow a simple fault-finding process.
4	Transmission, Braking & Steering	System diagrams, case studies, practical inspection	Case-study worksheet + practical checklist	Learner explains system purpose and performs basic inspection steps.
5	Diagnosis, Maintenance & Capstone	Fault scenarios, group activity, final practical exercise	Final quiz + practical assessment + capstone	Learner applies a structured approach to basic diagnosis and maintenance decisions.

4. Content Development Approach

Each module follows a consistent instructional sequence:

- Set induction / activation of prior knowledge
- Clear explanation of new concepts using simple language and diagrams
- Demonstration or worked example
- Guided learner activity
- Independent/practical application
- Formative assessment and feedback
- Recap and transition to the next topic

5. Sample Module Build — Engine System

Learning outcomes

- Identify the spark plug, piston, cylinder and crankshaft.
- State the basic function of each component.
- Describe, at an introductory level, how combustion produces mechanical movement.
- Use a simple checklist to inspect selected engine components safely.

Learning activities

- Display a labeled engine diagram and ask learners to identify familiar parts.
- Explain each component using a real or model engine where available.
- Demonstrate a basic visual inspection procedure.
- Give learners a guided component-identification exercise.
- Use a short fault scenario to encourage diagnostic reasoning.

Formative assessment

1. What is the function of a piston?
2. What does a spark plug do in a petrol engine?
3. What is the role of the crankshaft?
4. Name two safety precautions to observe before inspecting an engine.

6. Assessment & Rubric

Criterion	Excellent	Competent	Needs Improvement
Technical understanding	Accurately explains concepts and relationships	Explains main concepts with minor gaps	Shows major gaps or confusion
Practical execution	Completes task safely and independently	Completes task with limited guidance	Requires substantial guidance
Problem solving	Uses a logical diagnostic process	Follows a basic troubleshooting sequence	Approach is inconsistent
Communication	Clear, structured and technically appropriate	Generally clear	Unclear or incomplete

7. Capstone / Final Assessment

Learners receive a simple vehicle fault scenario. They must identify the relevant system, state the likely causes, propose inspection steps, observe workshop safety requirements, and communicate their recommended next action. The facilitator scores the task using the rubric above.

8. Completion Criteria

- Attend/complete the required learning sessions.
- Achieve the minimum passing score on the final knowledge assessment.
- Complete the required practical activities safely.
- Demonstrate the ability to explain core automotive systems and follow a basic diagnostic process.

9. Design Rationale

The program uses a progression from foundational knowledge to guided practice and then independent application. Learning objectives are measurable, activities are aligned with those objectives, and assessment is embedded throughout the program rather than being limited to the end. The structure supports different learner needs through explanation, visual material, demonstration, practice, feedback and practical assessment.

10. Portfolio Note

This document is presented as portfolio evidence of instructional program-design capability. It demonstrates the ability to translate a subject area into a complete learning pathway with objectives, modules, activities, educational content, assessments and completion criteria.